

Cloud Architecture & Deployment Standards

1. Purpose and Scope

This document defines the mandatory architecture and deployment standards for all production services operated by the Platform Engineering organization. It applies to every service deployed to the shared Kubernetes clusters, regardless of the owning team.

The standards cover service topology, deployment strategy, autoscaling policy, secrets management, and disaster recovery expectations. Teams that require an exception must file an architecture waiver with the Platform Review Board before launch.

2. Service Topology

Services are organized into three tiers: edge (public-facing gateways and load balancers), application (stateless business-logic services), and data (databases, caches, and message queues).

Application-tier services must be stateless and horizontally scalable; any local disk usage is treated as ephemeral and must not hold data required for correctness.

Each service must expose a `/healthz` endpoint for liveness and a `/readyz` endpoint for readiness. Readiness checks must verify connectivity to all hard dependencies, including the primary database and any required message broker.

3. Deployment Strategy

All production deployments use a rolling update strategy with a maximum surge of 25% and maximum unavailability of 0%. Canary releases are required for any service handling more than 1,000 requests per minute; canaries must run for at least 15 minutes and automatically roll back if the error rate exceeds 2% or p99 latency regresses by more than 30%.

Rollbacks must be executable within 5 minutes via a single command or pipeline trigger. Blue/green deployments are permitted for services with strict zero-downtime requirements, such as the payments gateway.

4. Autoscaling Policy

Horizontal Pod Autoscalers must be configured for every application-tier deployment, with a minimum of 3 replicas in production for high availability across zones. Scaling targets should be based on CPU utilization (70% target) and, where available, custom request-rate metrics.

Environment	Min Replicas	Max Replicas	Target CPU
dev	1	3	80%
staging	2	6	75%
production	3	50	70%

5. Secrets Management

Secrets must never be stored in source control, container images, or environment variable defaults checked into a repository. All secrets are provisioned via the central secrets manager and injected at runtime as mounted files or environment variables through the sidecar injector. Secrets are rotated automatically every 90 days for database credentials and every 30 days for API keys with external exposure.

6. Disaster Recovery

Every stateful data store must have a documented Recovery Point Objective (RPO) and Recovery Time Objective (RTO). The default RPO is 15 minutes (via continuous WAL archiving for relational databases) and the default RTO is 60 minutes. Disaster recovery drills must be executed at least twice per year, with results logged in the DR runbook.